

Statistics and AI driver's license

Using AI (safely) in practice: for learning and for data analysis

Hyungju Jeon · 2026/10/07

Introduction

Using AI in learning/research

What do you want to achieve through a PhD?

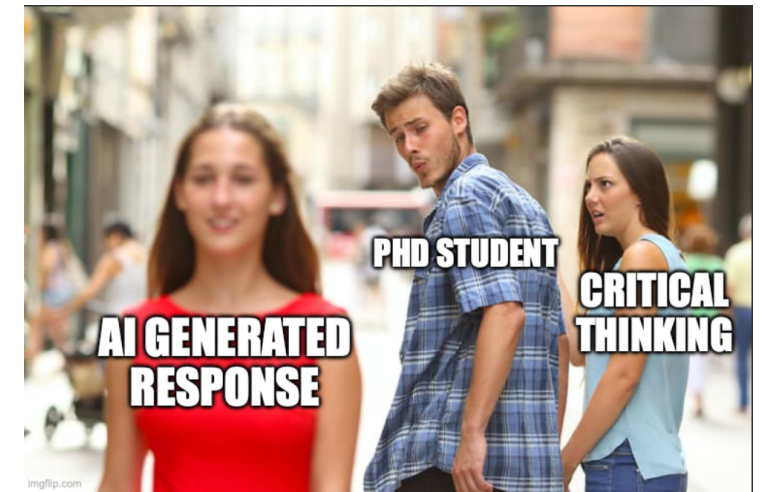
My answer: building a 'taste'. Knowing which questions matter, which results are interesting, and which analyses you can trust.

How can AI help you in achieving this goal?

Understand their strength and weakness, acknowledging risk

The temptation of productivity

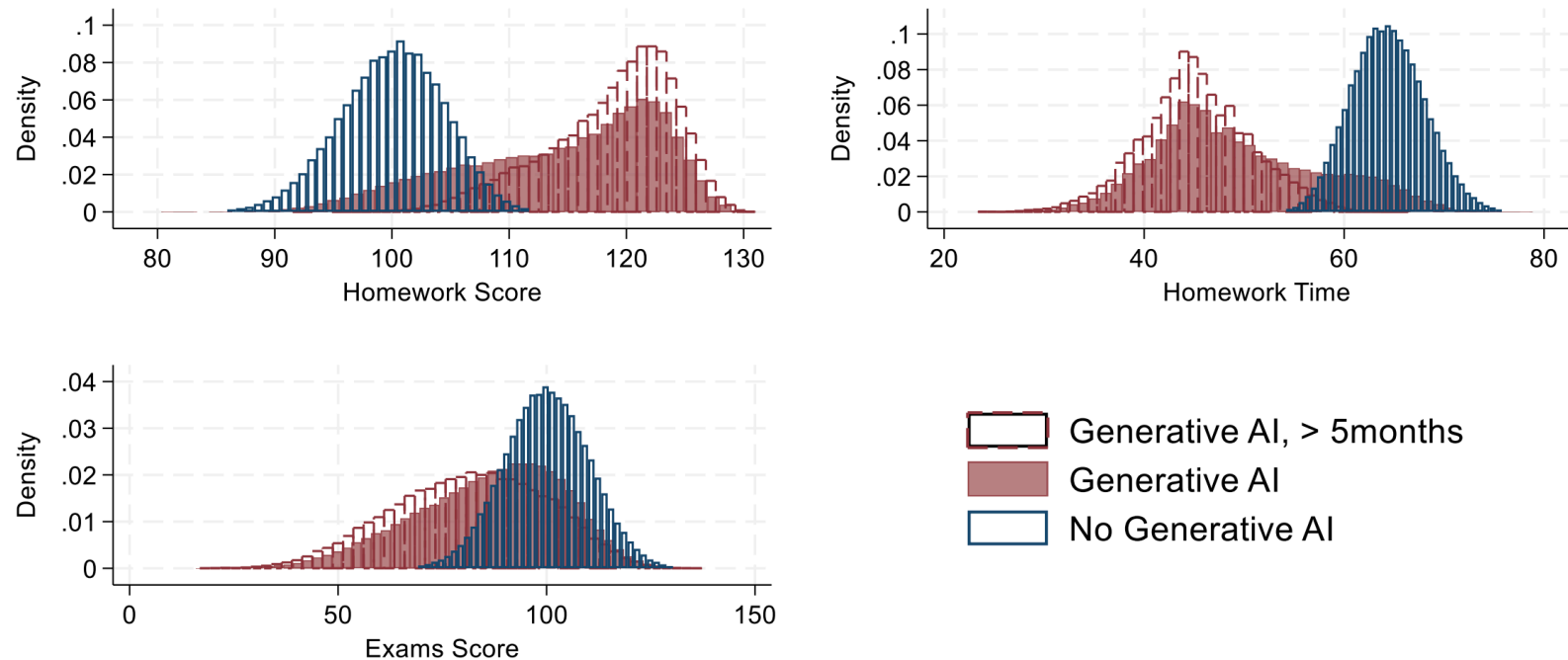
AI lets you produce more, faster: more code, more text, more analyses. It feels like progress, but if the AI does the thinking, you do not build your taste.



Illusion of competence

Reading and understanding AI response $\neq$ Learning

Histograms, GenAI > 5 months



Introduction

Illusion of competence

PNAS

RESEARCH ARTICLE

COMPUTER SCIENCES

OPEN ACCESS



The more you automate, the less you see: Hidden pitfalls of autonomous AI scientists

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Edited by Kenneth W. Wachter, University of California Berkeley, Berkeley, CA; received March 23, 2026; accepted September 2, 2026

The rapid rise of autonomous AI scientists marks a paradigm shift in scientific discovery by automating the research lifecycle. Yet their rushed development has outpaced critical oversight, leaving key workflow decisions dangerously unscrutinized. We present a much-needed systematic analysis of open-source AI scientist systems, investigating four primary pitfalls: inappropriate benchmark selection, data leakage, metric misuse, and post hoc selection bias. Through controlled experiments that isolate each pitfall, we find systematic vulnerabilities across two representative open-source systems. Crucially, we find that these flaws are largely invisible at the level of the final manuscript, suggesting that current manuscript-centric peer review paradigms are fundamentally insufficient for ensuring the integrity of automated research. We further propose mitigation strategies and demonstrate that access to full workflow artifacts (log traces and code) enables more effective auditing. Our findings suggest that journals, conferences, and researchers should move beyond manuscript-only evaluation toward process auditing the end-to-end workflow artifacts of AI scientist systems.

Significance

AI systems are increasingly capable of autonomously conducting scientific research, from generating hypotheses to writing full papers. However, our study finds that these “autonomous AI scientists” can introduce systematic methodological pitfalls such as biased evaluation, undisclosed data manipulation, and selective

[Submitted on 29 Aug 2025]

AI Meets the Learning?

Matthias Lehmann, Philipp B. Cornelius, Fabian J. Sting

The effect of large language models (LLMs) in education is debated: Previous research shows that LLMs can help as well as hurt learning. In two pre-registered and incentivized laboratory experiments, we find no effect of LLMs on overall learning outcomes. In exploratory analyses and a field study, we provide evidence that the effect of LLMs on learning outcomes depends on usage behavior. Students who substitute some of their learning activities with LLMs (e.g., by generating solutions to exercises) increase the volume of topics they can learn about but decrease their understanding of each topic. Students who complement their learning activities with LLMs (e.g., by asking for explanations) do not increase topic volume but do increase their understanding. We also observe that LLMs widen the gap between students with low and high prior knowledge. While LLMs show great potential to improve learning, their use must be tailored to the educational context and students' needs.

RESEARCH ARTICLE

ECONOMIC SCIENCES



Generative AI without guardrails can harm learning: Evidence from high school mathematics

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Hamsa Brunskill, Stanford University, Stanford, CA; received November 3, 2024; accepted May 5, 2025 by Editorial Board Member

122 (26) e2422633122 | <https://doi.org/10.1073/pnas.2422633122>

The Generative AI Learning Penalty: Evidence from Chinese Secondary Education

58 Pages • Posted: 23 Jun 2026 • Last revised: 3 Jun 2026

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AI makes you smarter but none the wiser: The disconnect between performance and metacognition

Published in:
Computers in Human Behavior

DOI:
[10.1016/j.chb.2025.108779](https://doi.org/10.1016/j.chb.2025.108779)

Using AI to learn

How can we use AI to learn?

Using AI to learn

Ways to learn with AI

Tailor the AI to your goals and to your learning style

Pattern	What it does
Tutor, don't solve	AI explains concept, and asks questions
Three levels	Explain a concept in multi-level abstraction
Interactive guide	Interactive visualization like 3Blue1Brown
AI as a student	Explain a concept in your own words; the AI asks questions and criticizes
AI Review	Point out weakness/errors in my claim but don't fix

And more...

Ask the same question twice

Chat 1: "What is the bootstrap?" Read the answer once.

Chat 2 : "Be my tutor for the bootstrap.

I know the sampling distribution of the mean and the shuffle test.

Do not give me the answer.

Ask me one question at a time, and wait for my answer."

Close both chats.

Q. What does the bootstrap resample, and what question does it answer that the shuffle test does not?

Prompt patterns

Pattern	Prompt
Tutor, don't solve	"Be my tutor for [topic]. Do not give me the answer. Ask me one question at a time, and wait for my answer."
Three levels	"Explain [concept] for a 5-year-old, an undergraduate, and a PhD student. Keep the facts the same. Tell me what you simplified."
Interactive guide	"Make one self-contained interactive HTML page with sliders that visualizes [concept] as I change [parameter]."
AI as a student	"I will explain [concept] in my own words. Act as a curious student: ask me questions until my explanation has no gaps."
AI Review	"Here is my [claim / code / summary]. Find the weakest point and any errors. Do not fix or rewrite it."

Reading code

Can you explain every line of the code?

Review your morning code

1 • Paste and run

Paste the test you wrote this morning into `code_reading.ipynb`.

2 • Ask for a review

“Do not rewrite or fix it. Say what null hypothesis it tests, then ask me questions.”

3 • Find a bug together

“I ran [this]. I expected [this], but got [that]. Do not fix it. Ask me one question.”

Let it quiz you

“Ask me 3 questions about this code, one at a time.”
Score yourself.

Check your code on a case you know

The coin

9 heads in 10 flips. Two-sided p from the worksheet table?

This morning's tumors

Lean mice, whole mm³: control 543, 83, 555, 483, 557; anti-PD-L1 0, 70, 0, 257, 29. 252 splits. Exact two-sided p ?

Predict first, then run your code.

AI version: change one line, predict, run

Change	Your prediction	Result
alternative="less", then "greater"		
Only 20 shuffles, five different seeds		
Remove the + 1 in the last line		

Write the prediction in your learning log before you run the cell.

If the AI's code is wrong, say exactly how: "I ran [this]. I expected [this], but got [that]. That is wrong because [reason]."

Explain it, then let the AI test you

Three levels — “Explain this code for a 5-year-old, an undergraduate, and a PhD student.”

A visual page — “Make one HTML page that animates one shuffle and builds the histogram.”

Quiz me — the AI asks 3 questions; answer without looking at the code.

AI-assisted learning in action

Different types of statistical errors

How often does the rule accuse an honest dealer?

This morning: a dealer flips a coin; you decide from 10 flips whether it is fair.

Your rule: watch 10 flips; accuse the dealer on 0, 1, 9, or 10 heads.

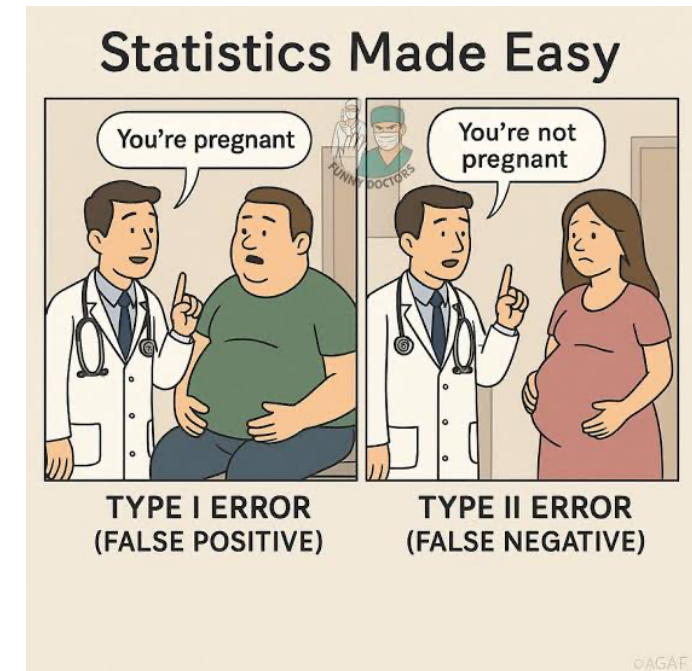
The dealer is honest. How often does your rule accuse them?

Heads in 10 fair flips	0	1	2	3	4	5	6	7	8	9	10
Probability	0.001	0.010	0.044	0.117	0.205	0.246	0.205	0.117	0.044	0.010	0.001

Two types of error

	Dealer is honest	Dealer cheats
Accuse	Type I	
Do not accuse		Type II

1. Which two cells are mistakes?
2. What is the mistake that we computed in the previous slide?
3. How can we compute the other error?



Catching one cheater: the 70% coin

Heads in 10 flips	0–2	3	4	5	6	7	8	9	10
Probability	0.002	0.009	0.037	0.103	0.200	0.267	0.233	0.121	0.028

1. How often does your rule (0, 1, 9, or 10 heads) accuse this cheater?
2. How often does this cheater get away?

Other cheating coins

Coin lands heads	50% (honest)	60%	70%	80%	90%
P(accuse)	0.022		0.15		

Compute the 60%, 80%, and 90% coins. Use $P(9 \text{ heads}) = 10 \cdot p^9 \cdot (1 - p)$ and $P(10 \text{ heads}) = p^{10}$; 0 and 1 heads are tiny.

1. Sketch $P(\text{accuse})$ against how often the coin lands heads. Where does the curve start?
2. Can the Type II error rate of your rule be one number? Write your idea.

AI step

“My rule: flip 10 times, accuse on 0, 1, 9 or 10 heads. The chance of catching a cheater is 0.048 (60% coin), 0.15 (70%), 0.38 (80%), 0.74 (90%). Can its Type II error rate be one number? My idea: [your answer to 2]. Do not give me the answer; ask me one question at a time.”

Reading the power curve

Coin lands heads	50%	60%	70%	80%	90%	100%
P(accuse), 10 flips	0.022	0.048	0.15	0.38	0.74	1

The class's power curve for your rule: 10 flips, accuse on 0, 1, 9, or 10 heads.

1. Where does the curve start, and why there?
2. How biased must the coin be before your rule catches it more often than not?
3. What new idea did the AI tutor give you on the last slide? Share one.

AI step:

“one HTML page that plots the power curve of my rule, with a slider for the number of flips.”

More flips, more power

Flips	Accuse when heads are	False accusations (α)	Catch the 70% cheater (power)
10	≤ 1 or ≥ 9	0.022	
20	≤ 5 or ≥ 15	0.041	
50	≤ 17 or ≥ 33	0.033	
100	≤ 39 or ≥ 61	0.035	

α stays below 0.05 at every size: it is set by the rule. Power grows with the number of flips.

The trade-off between α and β

Accuse when heads are	False accusations (α)	Catch the 70% cheater
0 or 10	0.002	
0, 1, 9 or 10	0.022	
0–2 or 8–10	0.109	

Moving the rule trades α against β .

How can we lower both?

AI step:

“one HTML page with two bar charts on one axis: heads in 10 flips for a fair coin and for a 70% coin. Let me move the rejection region, shade α and β , and show both numbers.”

Explain Type I and Type II errors

In your own words, with the dealer's coin: Type I error, Type II error, the power curve, and the trade-off between α and β .

1. Explain it to the AI.

"I will explain Type I and Type II errors. Do not reply until I write DONE. Then tell me what is right, what is missing, and what is wrong, and ask me one question about the weakest part."

2. Fix the weakest part. Then explain it to your neighbor without notes.

How many accused dealers are cheaters?

	Honest dealers (900)	Cheating dealers (100)
Accused		
Not accused		

In this casino, 1 in 10 dealers cheats with the 70% coin. You watch 10 flips of each of 1,000 dealers and use your rule.

1. Fill in the table. Of the accused dealers, what fraction cheat?
2. The manager says: “We accused 35 of 1,000 dealers. That is 3.5%, less than 0.05, so this casino has no cheating problem.” Is the manager right?
3. Redo 1 with 100 flips per dealer (accuse on ≤ 39 or ≥ 61 heads: $\alpha = 0.035$, power = 0.98), and again for a casino where half the dealers cheat. Which change raises the fraction more?

Estimating the cheat from caught cheaters

Flips	Accuse when heads are	Power (catch the 70% coin)	Estimated heads rate of caught cheaters
10	≤ 1 or ≥ 9	0.15	
20	≤ 5 or ≥ 15	0.42	
100	≤ 39 or ≥ 61	0.98	

All cheaters use the 70% coin. Estimate each caught cheater's coin from its heads (9 of 10 $\rightarrow$ 0.9).

1. Fill in the 10-flip row (you catch only 9 or 10 heads: $P = 0.121, 0.028$).
2. The truth is 0.70. Do caught cheaters look more biased? Predict the other rows.
3. Lab: 5 mice per group, a significant effect. Is the reported effect too big?

Peeking: testing again after more flips

You accuse an honest dealer ...	Probability
after the first 10 flips (≤ 1 or ≥ 9 heads)	0.022
only after 10 more flips (all 20: ≤ 5 or ≥ 15 heads)	
in total	
if you test only once, after all 20 flips (≤ 5 or ≥ 15 heads)	0.041

Step 1: watch 10 flips; accuse on ≤ 1 or ≥ 9 heads. Step 2: if not accused, watch 10 more; accuse if all 20 flips show ≤ 5 or ≥ 15 heads.

1. The dealer is honest. Can the total chance of an accusation be below 0.022?
2. Add the first two rows. Compare with testing only once, after all 20 flips.
3. Lab: $p = 0.08$ with 5 mice per group, so you add 5 more and test again. Problem?

Power of the shuffle test with few mice

Mice per group	Mice in total	Ways to split	Smallest p (two-sided)	Can it reach 0.05?
2	4	6		
3	6	20		
4	8	70		
5	10	252		

Two groups, drug and control. A shuffle test's $p = \text{splits at least as extreme} \div \text{all splits}$.

1. Compute the smallest possible two-sided p for each row. Hint: the best case for the drug.
2. With 3 mice per group, what is the power, however strong the drug?
3. How many mice per group before $p < 0.05$ is even possible? Like which coin rule?

Lean vs obese mice: similar effect, different p

This morning's tumors (5 vs 5 mice)	Difference in means	Spread of shuffled differences (SD)	Two-sided p
Lean mice	373 mm ³	158	0.016
Obese mice	295 mm ³	712	0.68

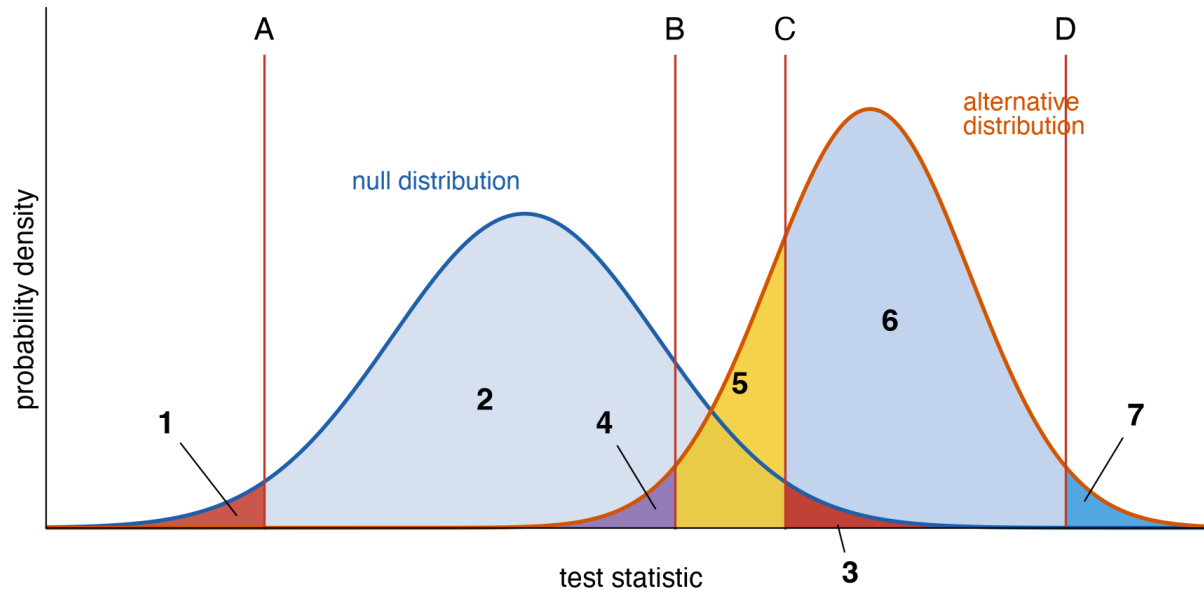
Same antibody, same experiment, 5 mice per group: only the kind of mouse differs.

1. Does $p = 0.68$ show that the antibody does nothing in obese mice?
2. The obese effect is 80% of the lean one, but 4.5 times noisier. What happens to the power?
3. To detect a 295 mm³ effect in obese mice, what would you change?

Quiz

Quiz! (Without AI)

Quiz: which area is which?



Lines A and C cut 2.5% tails of the null; B and D cut 2.5% tails of the alternative. $\alpha = 5\%$, two-sided.

1. Which areas add up to α ?
2. Which areas add up to β ?
3. Which areas add up to the power?
4. Which lines does the test use to decide?

Quiz: true or false

True or false? Give a one-line reason for each.

1. The p-value is the chance, if the null is true, of data at least as extreme as yours.
2. The p-value is the chance that the null hypothesis is true.
3. If $p > \alpha$, you can conclude that the null hypothesis is true.
4. Of two studies, the one with the smaller p-value found the larger effect.
5. For a fixed α and number of mice, β is the same whatever the size of the effect.
6. For the same true effect and the same α , more data gives more power.
7. If $p > 0.05$, adding mice and testing again keeps the false-positive rate at α .

Real data

Can AI analyse real data for you?

AI-assisted data analysis (1/2)

Data: BDNF in mouse cortex, Down syndrome model. 8 messy files. Open 06_real_data.ipynb.

0. Vibe research: upload the zip and ask “Analyse this data”. Keep its answer.
1. Clean: the AI writes code to merge the files. Check: 72 mice $\times$ 15 readings?
2. Look: the AI plots each mouse’s readings and mean by group. What is one observation?

AI-assisted data analysis (2/2)

3. Hypothesis: write yours first, AI closed: groups, null, one observation, test. Then the AI attacks your plan.

4. Test: the AI writes `my_test`. Check it on fake groups of saline mice: about 5% significant?

5. Show: the AI plots the result. You write two sentences; the AI attacks the weakest claim.

Compare with step 0. Where did the vibe answer and your analysis differ? Which do you trust?

Vibe research: same data, two answers

AI asked the plain question (pilot)	One observation =	p	Its conclusion
Claude Haiku (2 of 2 runs)	each reading (570 values)	0.003	memantine lowers BDNF
Claude Sonnet, Opus, GPT (6 of 6)	each mouse (38 means)	0.29	no evidence of a change

Your step 4 check: fake groups are significant 5.5% of the time with mouse means, 47% with readings.

1. Same data, same question: why two answers? Which one is right?
2. Without your check, how would you have known which answer to trust?

Optional • Multiple testing

What if you test 20 dealers?

If time allows.

Testing many honest dealers

Honest dealers tested	Chance that at least one is accused — your prediction
1	
5	
20	

Each honest dealer is accused with probability 0.022 (rule: 0, 1, 9, or 10 heads).

Hint: first find the chance that none is accused.

Design your own correction

In pairs, 5 minutes, AI closed.

Goal: test 20 dealers, and keep the chance of accusing any honest dealer at or below 0.05.

What does your procedure cost? Which cheaters will you now miss?

[Comic suggestion: xkcd 882 'Significant' — green jelly beans and acne.]

Bonferroni: a stricter rule for each dealer

Accuse a dealer when heads are	One honest dealer	Any of 20 honest dealers	Catch one 70% cheater
0, 1, 9 or 10	0.022	0.35	0.15
0 or 10	0.002	0.04	0.03

Dividing the allowed error by the number of tests ($0.05 / 20$) is the idea behind the Bonferroni correction. The price is power.

Then ask the AI to review your own procedure as a sceptic — without giving you its own.

Wrap-up

AI is a powerful tool, if you use it well

AI can be a great tutor.

Beware the illusion of productivity.

Always ask questions.

Find your own way to consolidate.

USES AI AS A TOOL



PRODUCTIVE PhD

COPY/PASTE AI RESPONSE



**DON'T UNDERSTAND
ANYTHING**

“The important thing is not to stop questioning. Curiosity has its own reason for existing.”

— Albert Einstein

Tip: quick side questions without losing your thread: /btw in Claude Code, side chat in Codex.